

SMART CITY EMERGENCY EVACUATION MAP

Python Programming Project

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Division: DIV 2 - Batch II

Subject: Python / Programming

Project: Smart City Emergency Evacuation Map using BFS

Code

```
# Smart City Emergency Evacuation Map
graph = {
    "MIT-ADT Campus": ["Kadamwak Wasti", "Loni Kalbhor"],
    "Kadamwak Wasti": ["Pune Station"],
    "Loni Kalbhor": ["Hadapsar"],
    "Pune Station": ["Shivaji Nagar Safe Zone"],
    "Hadapsar": ["Camp Area Safe Zone"],
    "Shivaji Nagar Safe Zone": [],
    "Camp Area Safe Zone": []
}

queue = []
visited = set()

start = "MIT-ADT Campus"

queue.append(start)
visited.add(start)

print("Checking Evacuation Route (BFS Traversal):")

while queue:
    node = queue.pop(0)
    print(" -> " + node)

    # Stop the search if we reach a goal (Safe Zone)
    if "Safe Zone" in node:
        print("\n■ Nearest Safe Zone Reached! Evacuation Successful.")
        break

    for neighbour in graph[node]:
        if neighbour not in visited:
            visited.add(neighbour)
            queue.append(neighbour)
```

Project Overview

This project demonstrates Breadth First Search (BFS) using a smart-city emergency evacuation map. The graph represents locations and possible evacuation routes. BFS starts from MIT-ADT Campus, explores routes level by level, and stops when a Safe Zone is reached.

How the Program Works

1. The graph stores each location and its connected destinations.
2. A queue is used to process locations in BFS order.
3. The visited set prevents the same location from being processed repeatedly.
4. The search begins at MIT-ADT Campus.
5. When a location containing 'Safe Zone' is reached, the program reports successful evacuation and stops.

Expected Traversal

MIT-ADT Campus → Kadamwak Wasti → Loni Kalbhor → Pune Station → Hadapsar → Shivaji Nagar
Safe Zone

Note

The BFS traversal checks locations in queue order. Because both branches are added from the starting location, the exact first Safe Zone reached depends on the order in which neighbours were inserted into the graph.